Experiment 11

Study Dynamic Response of Single Machine Infinite Bus (SMIB) System

Lab Report

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Section B

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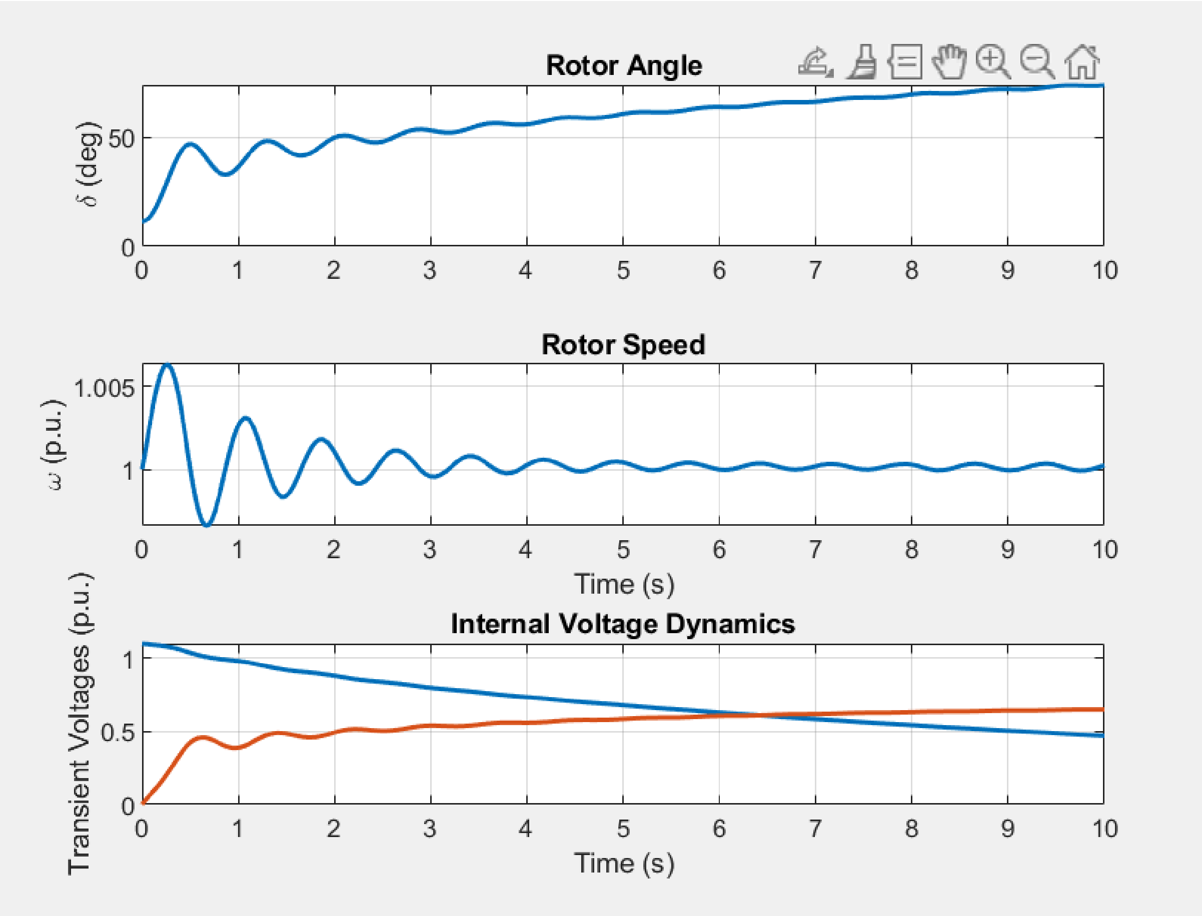
MATLAB code with clear comments:

```
%% Experiment: Single Machine Infinite Bus (SMIB)
clear; clc; close all;
f = 50;
ws = 2*pi*f;
H = 6.175;
D = 0;                          % Damp
Xd  = 1.8;
Xq  = 1.7;
Xd_p = 0.3;
Xq_p = 0.55;
Tdo_p = 8.0;
Tqo_p = 0.4;
```

```
V = 1.0;
theta = 0;
Pm = 0.8;                   % Mechanical power input (p.u.)
Q  = 0.2;                   % Reactive power (p.u.)
% x = [delta omega Eqp Edp]
delta0 = 0.2;    %
omega0 = 1.0;    % Rotor speed
Eqp0  = 1.1;     % q-axis transient EMF
Edp0  = 0.0;     % d-axis transient EMF
Efd = 1.2;
%% Initial state vector
% x = [delta omega Eqp Edp]
x0 = [delta0; 1; Eqp0; Edp0];
%% Simulation
tspan = [0 10];
[t, x] = ode45(@(t,x) smib_dynamics(t,x, ...
   Pm,Efd,V,theta, ...
   H,D,Xd,Xq,Xd_p,Xq_p,Tdo_p,Tqo_p,ws), ...
   tspan, x0);
%% Plot results
figure;
subplot(3,1,1)
plot(t, rad2deg(x(:,1)),'LineWidth',1.5)
ylabel('\delta (deg)')
title('Rotor Angle')
grid on
subplot(3,1,2)
plot(t, x(:,2),'LineWidth',1.5)
ylabel('\omega (p.u.)')
xlabel('Time (s)')
title('Rotor Speed')
grid on
subplot(3,1,3)
```

```
plot(t, x(:,3),'LineWidth',1.5)
hold on;
plot(t, x(:,4),'LineWidth',1.5)
ylabel('Transient Voltages (p.u.)')
xlabel('Time (s)')
title('Internal Voltage Dynamics')
grid on;
%% --------------------------------------------------
%% Generator dynamics (SMIB)
function dx = smib_dynamics(~,x,Pm,Efd,V,theta,...
                            H,D,Xd,Xq,Xd_p,Xq_p,Tdo_p,Tqo_p,ws)
delta = x(1);
omega = x(2);
Eqp   = x(3);
Edp   = x(4);
% dq voltages
Vd = V*sin(delta - theta);
Vq = V*cos(delta - theta);
% dq currents
id = (Eqp - Vq)/Xd_p;
iq = (Vd - Edp)/Xq_p;
% Electrical power
Pe = Vd*id + Vq*iq;
dx = zeros(4,1);
% Swing equations
dx(1) = ws*(omega - 1);
dx(2) = (1/(2*H))*(Pm - Pe - D*(omega-1));
% Flux decay equations
dx(3) = (1/Tdo_p)*(Efd - Eqp - (Xd - Xd_p)*id);
dx(4) = (1/Tqo_p)*(-Edp + (Xq - Xq_p)*iq);
end
```

RESULTS



Rotor Angle
δ (deg)
0
50
0
1
2
3
4
5
6
7
8
9
10
Rotor Speed
ω (p.u.)
1.005
1
Time (s)
Internal Voltage Dynamics
Transient Voltages (p.u.)
0
0.5
1
Time (s)